

Lecture Unit 1.2

The Top500 List

Measuring Performance

- Performance matters for HPC but how do we measure performance?
- No single number can fully capture all aspects of performance
- There are many ways to quantify the performance of an HPC system or HPC application and choosing an appropriate metric depends what we are interested in
- We will encounter other performance metrics and benchmarks during this module

Linpack

- HPC applications typically perform many operations on floating point numbers (e.g. float/ double in C/C++)
- Linpack is a performance benchmark which measures **f**loating point **o**perations per second (flops) using a dense linear algebra workload
- A widely used performance benchmark for HPC systems is a parallel version of Linpack called HPL (High-Performance Linpack)

TOP500 List

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Call for participation in the 56th TOP00 / GREEN500 lists

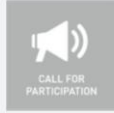
Oct. 13, 2020

The TOP500 authors encourage the HPC community to submit entries for the 56th TOP500 and Green500 lists.

The November 2020 TOP500 list is released during the SC20 Supercomputing Conference.

We are interested in new entries as well as entries that are no longer valid. Please feel free to contact the authors [here](#) if you have questions. The guidelines for submission of entries can be found [here](#).

[read more »](#)

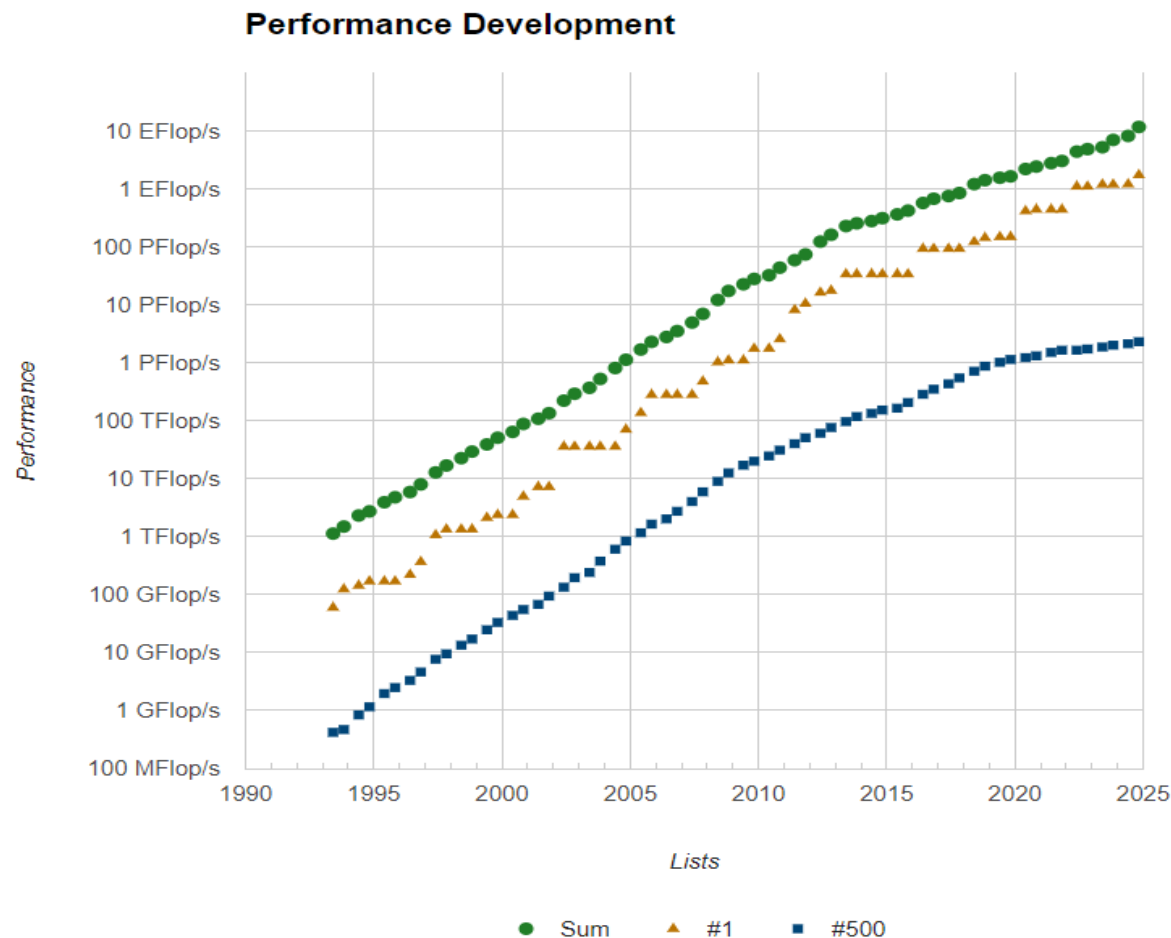


<https://www.top500.org>

- The fastest known machines in the world appear in the “Top 500” list
- The list is published twice a year and ranks HPC systems according to HPL performance
- The Top 500 is a long term data set which reveals trends in HPC since mid 1990’s

TOP500 Performance Development Over Time

- Performance has grown exponentially (log- linear plot)



Multipliers for flops

Name ↕	Unit ↕	Value ↕
kiloFLOPS	kFLOPS	10^3
megaFLOPS	MFLOPS	10^6
gigaFLOPS	GFLOPS	10^9
teraFLOPS	TFLOPS	10^{12}
petaFLOPS	PFLOPS	10^{15}
exaFLOPS	EFLOPS	10^{18}
zettaFLOPS	ZFLOPS	10^{21}
yottaFLOPS	YFLOPS	10^{24}
ronnaFLOPS	RFLOPS	10^{27}
quettaFLOPS	QFLOPS	10^{30}

<https://www.top500.org/statistics/perfdevel/>

November 2025 TOP5

Rank	System	Cores	Rmax (PFlop/s)	Rpeak (PFlop/s)	Power (kW)
1	El Capitan - HPE Cray EX255a, AMD 4th Gen EPYC 24C 1.8GHz, AMD Instinct MI300A, Slingshot-11, TOSS, HPE DOE/NNSA/LLNL United States	11,340,000	1,809.00	2,821.10	29,685
2	Frontier - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE Cray OS, HPE DOE/SC/Oak Ridge National Laboratory United States	9,066,176	1,353.00	2,055.72	24,607
3	Aurora - HPE Cray EX - Intel Exascale Compute Blade, Xeon CPU Max 9470 52C 2.4GHz, Intel Data Center GPU Max, Slingshot-11, Intel DOE/SC/Argonne National Laboratory United States	9,264,128	1,012.00	1,980.01	38,698
4	JUPITER Booster - BullSequana XH3000, GH Superchip 72C 3GHz, NVIDIA GH200 Superchip, Quad-Rail NVIDIA InfiniBand NDR200, RedHat Enterprise Linux, EVIDEN EuroHPC/FZJ Germany	4,801,344	1,000.00	1,226.28	15,794
5	Eagle - Microsoft NDv5, Xeon Platinum 8480C 48C 2GHz, NVIDIA H100, NVIDIA Infiniband NDR, Microsoft Azure United States	2,073,600	561.20	846.84	

Unit Summary

- Linpack measures performance (flops) using a dense linear algebra workload
- The Top500 list ranks HPC systems according to Linpack (HPL) performance
- The Top500 is a useful indicator of global trends in HPC and we will refer to it again later in the module
- Today the fastest systems in the world achieve 10's-100's petaflops (10^{15} flops) performance using over 1 million processor cores